

PHY F376: A study of hybrid classical-quantum algorithms

Week 3 (3/9/2026) - Implementing a Real Molecular Hamiltonian: The H₂O CAS(4e,4o) Qubit Hamiltonian
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Overview of This Week's Work

This week I moved from the N₂ tutorial example to an actual Hamiltonian provided for the project: an 8-qubit Jordan-Wigner-transformed Hamiltonian for water, obtained from a CASSCF calculation with an active space of 4 electrons in 4 spatial orbitals. Unlike the previous weeks, this Hamiltonian was given directly as a list of Pauli terms and coefficients rather than something built from scratch through PySCF, so the task was to convert this file into a usable quantum-computing object and confirm that it was parsed correctly.

Converting the Pauli List into Code

The file lists one Pauli term per line, each qubit's operator written with its index attached (for example `Y0 Z1 Y2`), with any qubit not mentioned implicitly assumed to be identity. I wrote a parser that reads each line, builds an 8-character Pauli string for that term, and reverses it before use, since Qiskit's convention reads a Pauli string right to left with qubit 0 as the rightmost character. Each parsed line becomes one `(label, coefficient)` pair, and collecting all of them into `SparsePauliOp.from_list(...)` builds the full Hamiltonian as a single weighted sum of Pauli strings, exactly matching the qubit-operator form of the molecular Hamiltonian discussed in previous weeks.

Verifying the Result

Since this Hamiltonian only acts on 8 qubits, I could check the parsing by brute force: converting it to its full 256×256 matrix, confirming it was Hermitian, and diagonalizing it exactly with NumPy. The lowest eigenvalue matched the CASSCF reference energy stated in the file to about ten decimal places, which confirmed both that the parsing logic was correct and that the qubit-ordering convention had been handled properly. This kind of full diagonalization is of course only possible because the system is small; it would not scale to the systems SQD is actually meant for, but it was a useful way to validate the code before trusting it further.

Making Sense of the Terms Physically

Going through this Hamiltonian also clarified what the individual Pauli terms mean physically, which I had found somewhat abstract before. Since each qubit corresponds to one spin-orbital, single- and two-qubit Z terms are diagonal and simply raise or lower the energy depending on whether specific orbitals are occupied, without changing the electron configuration at all. Terms built from X and Y , on the other hand, flip qubits and therefore genuinely move electrons between orbitals; these always appear in pairs or larger combinations so that the total electron count stays fixed, and it is exactly these terms that allow the true ground state to become a mixture of several electron configurations rather than a single fixed one.

Concluding Remarks

This week's exercise made the Hamiltonian feel far less abstract: rather than a formula, it is now something I have built, checked, and can reason about term by term. One open point to resolve before this Hamiltonian can be used with the existing LUCJ-based workflow is that its qubit ordering (alpha and beta orbitals interleaved) differs from the block ordering assumed in the N₂ example, and this will need to be addressed before proceeding. With a verified Hamiltonian in hand, I am ready to begin working on actually running a quantum workflow on it.